

Centralized Booking System for Pet Grooming in Makati: Serving Owners and Groomers

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School of Computing and Information Technologies

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Systems Analysis & Detailed Design for IT-MI

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Introduction

Pet ownership in the Philippines is rapidly growing with approximately 16.4 million pet dogs alone in 2020 [1]. The pandemic significantly contributed to this increase due to lockdown measures and heightened need for companionship. An informal survey revealed that 83% of Filipinos consider their pet their "baby" and 8% their sibling, establishing them as essential family members and budget considerations [2]. Consequently, the pet industry has expanded across pet food and accessory manufacturing, services including grooming and accommodation, and pet-friendly establishments meeting this growing demand. While pet owners require convenient access to reliable pet services, particularly grooming and accommodation, the current market exhibits significant inefficiencies. Pet owners struggle with service provider comparison and appointment scheduling challenges, while service providers lack organized booking systems and remain dispersed across social media platforms. This proposal aims to understand the booking preferences of pet service providers and pet owners to provide a convenient and hassle-free booking experience.

1. Pet Service: Inefficient Booking System. Service providers commonly lack structured booking systems, resulting in scheduling conflicts through predominantly manual processes conducted via social media messaging [3]. The Philippine Veterinary Medical Association (PVMA) has documented a nationwide veterinarian shortage [4], exacerbating difficulties for pet owners attempting to secure appointments due to limited availability of slots and pet groomers.
2. Pet Service: Dispersed Digital Market Landscape. Small service providers suffer from limited visibility due to their dispersion across social media without a centralized platform. Alex emphasized that the right website could transform small businesses by showcasing services continuously, expanding customer reach, and enhancing competitive positioning [5]. Furthermore, pet owners face challenges comparing service providers amid scattered information and absence of standardized evaluation metrics for quality, pricing, and availability [6].

The pet service industry faces two interrelated challenges that affect both service providers and pet owners. First, the inefficient booking systems prevalent among small providers create scheduling conflicts and appointment difficulties. Second, the existing process of listing services results in inefficient comparison of services, with providers scattered across various platforms

without standardized metrics for evaluation. These challenges underscore the need for a solution that addresses both the organizational requirements of service providers and the information accessibility needs of pet owners in this rapidly growing market.

Project Context

As a startup, *furlink* aims to modernize the way pet grooming services are booked and managed in the Philippines. Unlike platforms such as Agoda or Airbnb—which focus on property and hotel bookings—*furlink* specializes in connecting dog or cat owners with pet grooming service providers through a streamlined digital platform.

furlink offers a hassle-free experience for both customers and service providers. It enables grooming businesses to advertise their services while allowing pet owners to discover, schedule, and manage appointments in one place. Additionally, the platform integrates an innovative AI-powered grooming preview tool, giving users a visual reference of potential pet haircut styles before booking—bridging the gap between customer expectations and actual grooming outcomes.

Designed as a local-first startup solution, *furlink* initially targets grooming services within select cities, with plans to expand as adoption grows.

Statement of the Problem

The following problems of pet owners and service providers in Makati City were identified through a conducted survey which we aim to address by providing a website platform. Data was gathered from two (2) separate sets of questionnaires for pet owners and pet grooming service providers. Among the sixty-four (64) respondents from the survey conducted for pet owners, forty-three (43) were identified as pet owners from Makati City.

The forty-three (43) responses revealed that the most frequently reported issue among pet owners was booking related issues, ranging from double booking to unorganized scheduling systems. On the other hand, nine (9) respondents answered issues related to unmet pet haircut expectations.

1. Lack of Ability to Customize Pet Haircuts

Pet owners often do not know what grooming styles are suitable for their dog or cat, leading to unmet expectations and miscommunication with groomers. Many owners are also unaware of the full range of grooming options available for their specific breed or coat type. The absence of an effective haircut preview tool results in hesitant client decisions and missed opportunities for salons to enhance their digital offerings and client engagement.

2. Conflict in Schedules

Current common scheduling methods such as contacting pet service providers through social media platforms frequently result in double bookings, missed appointments, and inefficient time management for service providers. This not only frustrates providers but also inconveniences pet owners, leading to lost revenue and poor customer experience.

3. Low Provider Visibility

Small to medium-sized pet grooming service providers often struggle with limited visibility due to their physical location, making it difficult for pet owners to discover their services. These businesses are frequently placed in less prominent areas, causing them to be overshadowed by larger, more established service providers. This significantly hinders their ability to expand their market and gain more customers.

4. Lack of Business Reports

Many independent pet service providers lack access to data on their performance, sales, and customer feedback. This absence of actionable insights makes it difficult for them to track growth, understand customer satisfaction, and make informed business decisions to improve their services.

Objectives

Main objective: To provide a better and simpler solution for dog or cat owners to efficiently find and book dog or cat grooming services while also helping pet service providers to manage their services.

Specific objectives:

1. Ability to Customize Pet Haircut

Successfully develop and deploy an AI-powered haircut preview tool, integrated into the platform, by September 6, 2025, ensuring it achieves a minimum of 70% image rendering accuracy, accurate identification of selected haircut suitability for the pet's current hair style and condition.

2. Reduce Scheduling Conflicts

Minimize double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release.

3. Improve Visibility of Pet Service Providers

Improve the visibility of small to medium-sized pet service providers by giving them a dedicated platform to showcase their services and directly connect with pet owners. Onboard and verify one (1) service provider within the first three (3) months of release, and each provider receives search engine optimization (SEO) profile dashboard to improve visibility and potential customer reach.

4. Provide Business Insights for Service Providers

Enable pet service providers to track monthly sales, customer feedback, and related insights by integrating a business analytics dashboard into the platform by January 8, 2025.

Project Scope

This proposed project aims to develop a website platform to connect dog or cat owners with dog or cat grooming service providers in Makati City, creating more efficient access to pet grooming services. Meanwhile, it also allows service providers to have a wider and a more inclusive reach to their potential customers and organization of their booking system. As a result, this proposed project aims to increase the economic growth through amplifying the pet service industry, making this proposal aligned with SDG number 8.

This project will include the following features:

1. User registration and login: Allows both dog or cat owners and service providers to create an account and login to their account.
2. Service listing: Allows service providers to post their services on the platform.
3. Booking: Enables dog or cat owners to book pet services on the platform.
4. AI pet haircut generator: Enables dog or cat owners to upload a photo of their pet and receive an AI-generated preview for their desired haircut style.
5. Messaging channel: Allows dog or cat owners and pet service providers to communicate with each other.
6. Pet service provider recommendation: Provides dog or cat owners recommended pet service providers based on their activities.
7. Dashboard: Allows pet service providers to track bookings, revenue, and customer insights.
8. Feedback form: Allows dog or cat owners to submit their review to the service provider.

The project includes dog or cat owners in Makati City that seeks new service options and are open to digital solutions. Another inclusion are small to medium-sized dog or cat grooming service providers based in Makati City that are not located in well-known areas.

This project will exclude the following features:

1. Booking of pet care services with insurance.
2. Other pet services such as veterinary care, pet supplies, etc.
3. Pet transportation services

4. The system does not process any payments. It does not use in-app payment through third-party gateways.
5. In-depth financial reporting tools.

Success Criteria

How you determine that the project and deliverables have been completed successfully.

- Unit Testing
 - Entry Criteria: Individual functions or modules have been developed, and code has undergone peer review and is ready for testing.
 - Exit Criteria: 100% of unit test cases pass, no critical or major bugs, and code is at least at 80% completion.
- Integration Testing
 - Entry Criteria: Key modules (e.g., user registration/login, booking flow, service listing, AI haircut generator, calendar sync, dashboards) have passed unit testing, and modules are connected and ready for end-to-end flow testing.
 - Exit Criteria: All integration test cases are successfully executed wherein all data are synchronized and correct between modules and priority features are done.
- System Testing
 - Entry Criteria: All modules have been integrated into the staging environment and functional requirements are on freeze.
 - Exit Criteria: 95% of system test cases passed, the system works as expected and no critical bugs remain.
- User Acceptance Testing
 - Entry Criteria: System is complete, stable and ready to be tested by test users or stakeholders.

Exit Criteria: Test users are done with testing the system and critical UAT feedback is addressed, and the system is ready for deployment.

Project Assumptions

- If there is a potential client that does not have a fully working system.

- If there is a potential client, they will fully fund the financial requirements of the project.

There will be one pet grooming service provider that will onboard.

Project Constraints

- Limited financial resources.
- Heavily relies on open-source and free-tier tools.
- Limited time given.

Project aims to release its MVP by October 18, 2025 and completed by January 17, 2026.

Project Charter

Purpose

The purpose of this project is to modernize the way pet grooming services are booked and managed in Makati City. It aims to reduce schedule conflicts, miscommunication, enhance customer satisfaction, and improve service visibility for small to medium-sized pet grooming shops by providing tools like AI haircut preview generator, clear booking process, and business insights.

High-level project description

furlink is a web-based booking system designed to help dog or cat owners to discover pet groomers in Makati City, provide tools to manage bookings, provide feedback, and include features such as AI pet haircut preview generator, business insights for pet service provider, and calendar-based scheduling. The project will cater to dog or cat owners and pet grooming service providers in Makati City that seek new service options and are open to digital solutions.

High-level milestone schedule

October 9, 2025 – Completion of MVP

October 18, 2025 – furlink v1.0 Release (MVP Release)

December 4, 2025 – furlink v1.1 Release (Feedback and messaging channel)

January 8, 2026 – furlink v2.0 Release (Dashboard and Help Center)

Rough cost estimate and budget

Table 1 Estimated Budget

Item/Service	Unit Cost in PHP	Duration	Total in PHP
Vercel Pro Plan	~1,200 /month	6 months	7,200
Supabase (Postgres)	~ 1,000 /month	6 months	6,000
Firebase Firestore + Authentication	~ 1,500 /month	6 months	9,000
AWS S3	~1,200 /month	6 months	7,200
Domain registration	~2000 /year	1 year	2,000
Cloud Monitoring / Logging (basic tools free tier)	0	-	0
Total	31,400		

Stakeholders

- Pet Owners
- The Groom Room by Steve
- The Nature Pet Grooming Salon
- LogiTech members
- Project Adviser
- Project Consultants
- Competitors

Project manager

Name: Lalainne C. Andaya

Position: Project Manager, Backend Developer

Project manager's responsibilities

- It is the project manager's responsibility to:
- Coordinate project activities and ensure all milestones and deadlines are met
- Communicate progress and blockers to stakeholders
- Monitor documentation and ensure all features align with the project objectives

Project manager's authority

- The project manager has the authority to:
- Approve task division and internal team decisions
- Represent the team during presentations and meetings
- Coordinate directly with stakeholders

Formal declaration of sponsor's support

Our project, Furlink, is a student-led startup and our primary sponsors consist of the project team members themselves. The team is committed to contribute time, effort, and adequate resources to ensure that the project will be successful. Additionally, the members' immediate family members are willing to provide needed support to drive the success of furlink. While they are not directly involved in decision-making they serve as sponsors by backing the project financially and offering moral and logistical support as needed.

Thus, the group formally acknowledges the role of their parents and personal contributions as the foundational sponsorship for this project. The collective support strengthens the team's ability to deliver a meaningful and functional project, even

To further strengthen our project, we are actively seeking and incorporating consultation from professionals, academic advisers, and our target users. This valuable input ensures our project is not only viable but also directly addresses the needs of our community, guiding us toward the best possible outcome.

Stakeholder Analysis

The stakeholder analysis helped identify furlink's stakeholders, their objectives, and their potential contributions to and resistance against the project.

Please click this link to view the Stakeholder Analysis [Stakeholder Analysis](#)

Open Project Activities

Please click this link to view the Openproject Activities [Openproject Activities](#)

Design Thinking Stages

Design Thinking is a human-centered problem-solving process with five stages: empathize with users, define the problem, generate ideas, create prototypes, and test solutions. It's iterative, allowing continuous improvement until the best solution is reached.

Please click this link to view the Design Thinking Stages [Design Thinking Stages](#)

Use Case Diagram

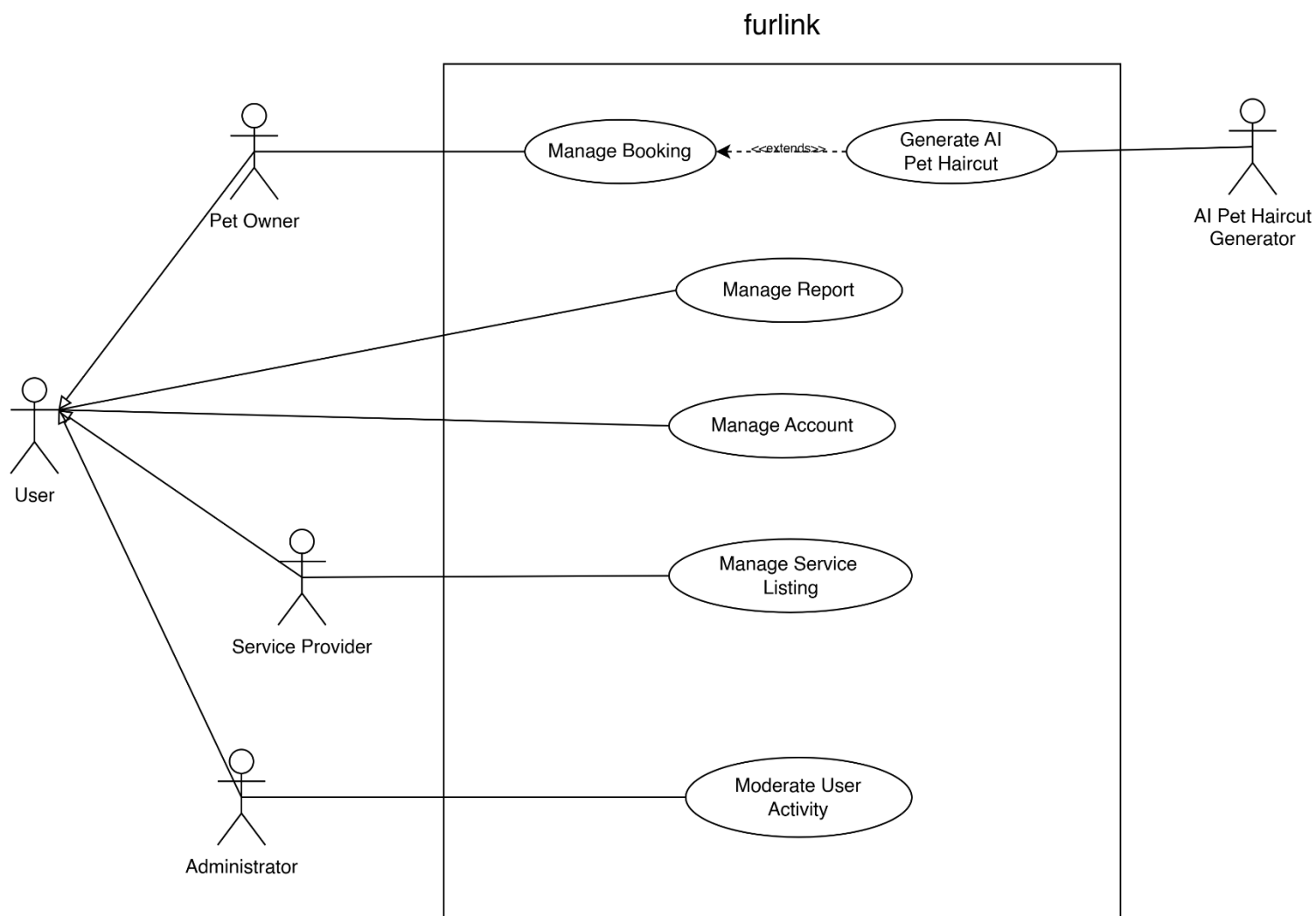


Figure 1 Use Case Diagram

Use Case Full Description

The purpose of this Use Case Full Description is to provide an overview of how users (dog or cat owners and pet grooming service providers) and the system administrator interact with the system.

Please click this link to view the use case full description of *furlink* [Use Case Full Description](#)

Data Flow Diagram

Context Diagram

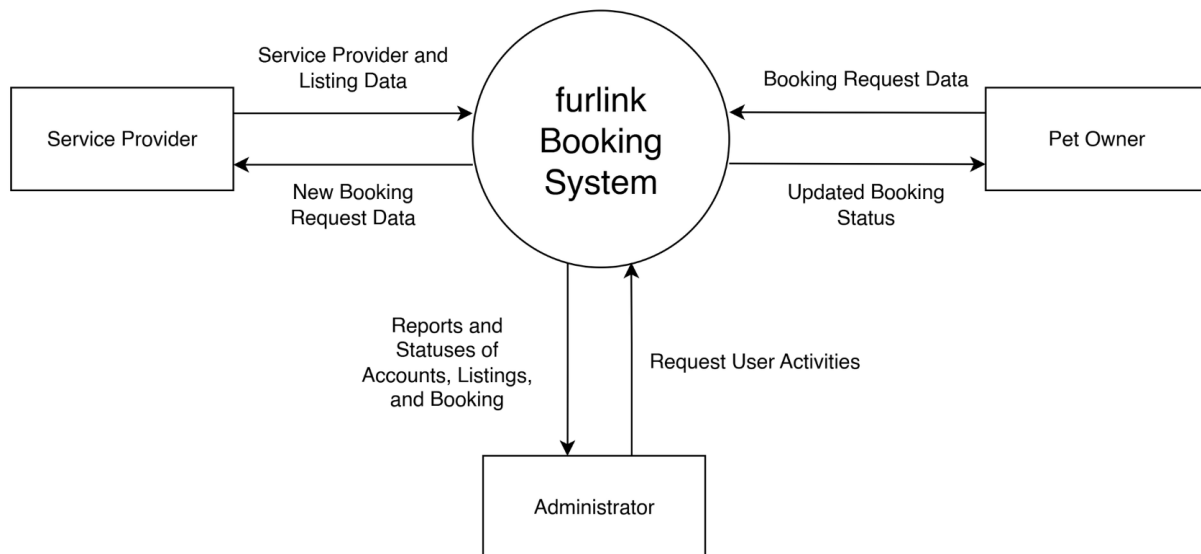


Figure 2 Context Diagram

The diagram above is for a booking system called *furlink* at high-level. It shows how the system interacts with external entities such as the service providers and the pet owners. Pet service providers can manage their listings by adding, viewing, updating, or deleting their services. They can also receive booking requests as initiated by pet owners. Meanwhile, pet owners enter their personal information along with details of their pets related to the grooming booking request. Once they request a booking appointment, pet owners receive confirmation of their scheduled appointment through the system.

On the other hand, a system administrator can request user activities. Once processed, the system will provide reports and statuses of accounts, listings, and bookings.

Level 1 Data Flow Diagram

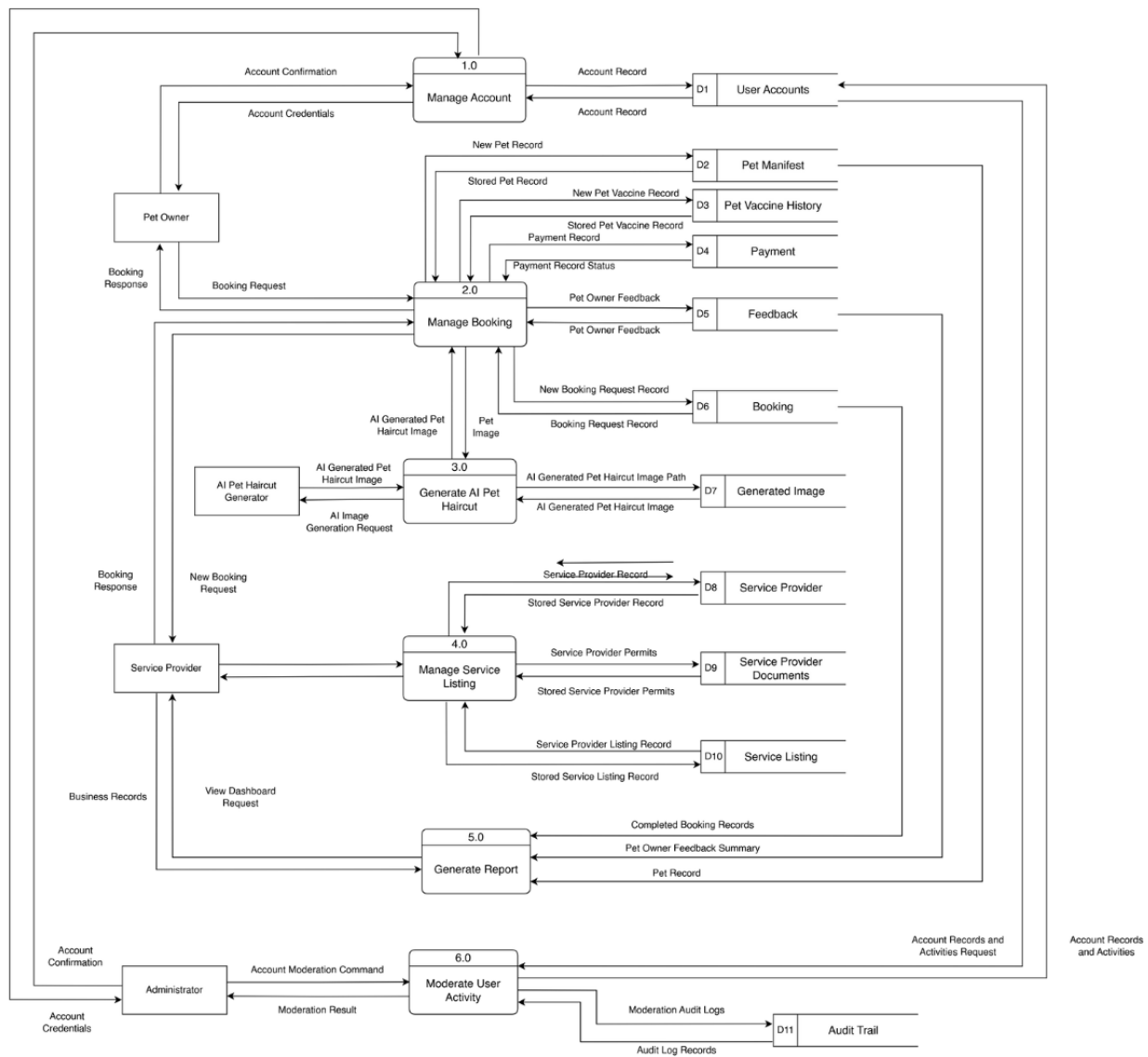


Figure 3 Level 1 Data Flow Diagram

The figure above is the Level 1 Data Flow Diagram which provides a more detailed view of furlink's system compared to the high-level Context Diagram. It shows the system's main processes, external entities, and data stores. A pet owner can Login or Signup through Process 1.0 and modify their account through Process 2.0, which interacts with the User Accounts data store. They can book services by sending a booking request with the pet and booking data which are stored in different data stores. To become a pet service provider, one must be logged in as a pet owner first and request to upgrade their access, so they are able to use service provider features. Once

a pet owner has upgraded their access, they can now post and manage their services as service providers. A system admin can only manage user activities once they have logged in and have their credentials authenticated.

Level 2 Data Flow Diagram: Manage Account

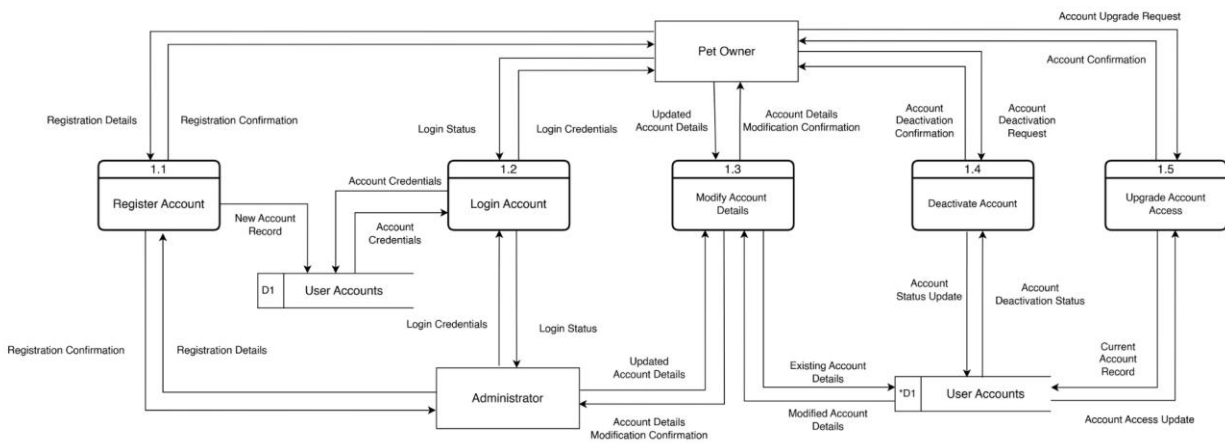


Figure 4 Level 2 Data Flow Diagram Manage Account

The figure above is the Level 2 diagram which shows an list of extended processes users can perform on the system such as account registration, login, modify, delete, and upgrading the pet owner account access to service provider.

Level 2 Data Flow Diagram: Manage Booking

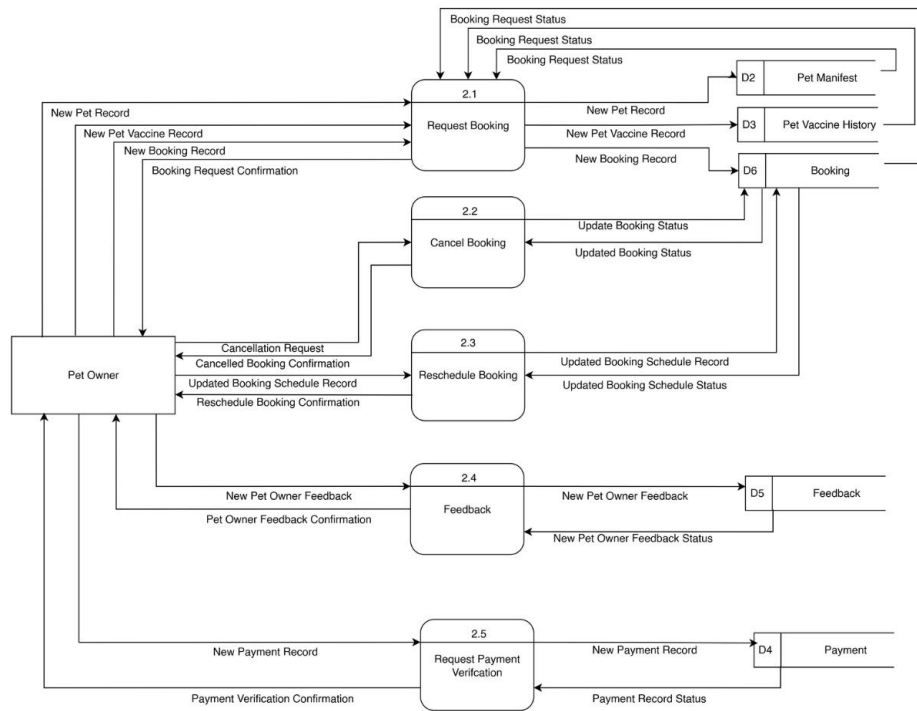


Figure 5 Level 2 Data Flow Diagram Manage Booking

The DFD Level 2 for Manage Booking details the interactions that occur when a pet owner uses the Furlink platform to book a service. The process begins with the Request Booking activity, where the pet owner completes the booking form, providing a new pet record, vaccine record, and booking details. This information is then stored in the Pet Manifest, Pet Vaccine History, and Booking Data Store. Upon submission, the pet owner receives a booking request confirmation, which includes the booking's status.

Level 2 Data Flow Diagram: AI Per Haircut Generator

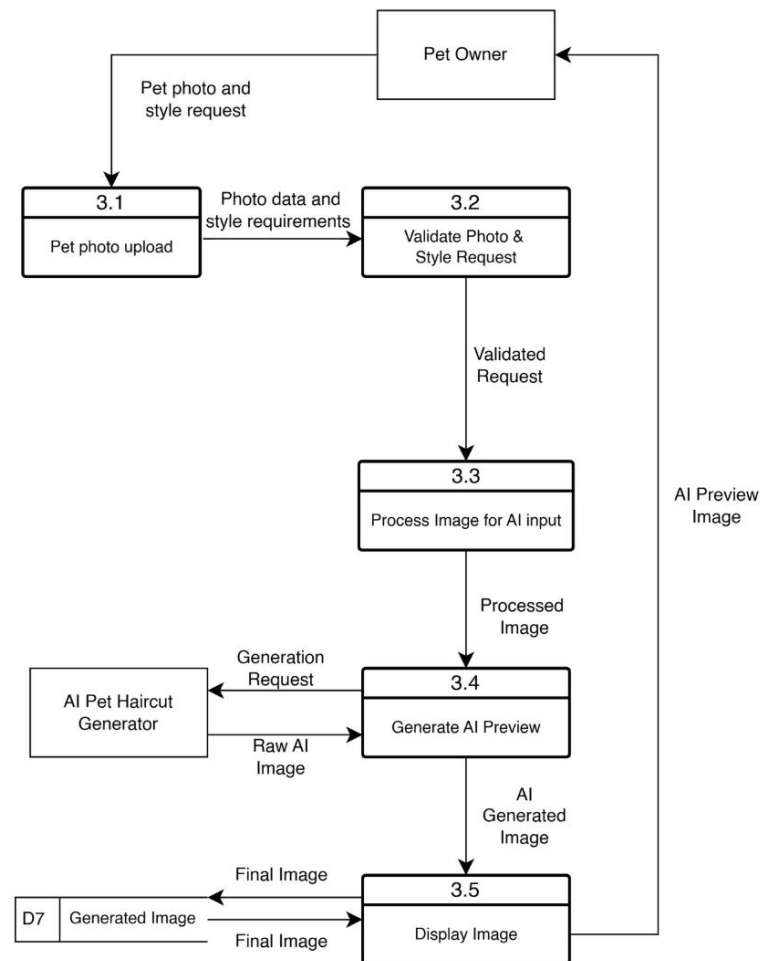


Figure 6 Level 2 Data Flow AI Pet Haircut Generator

For the level 2 DFD of our AI pet haircut generation, the pet owner will upload their pet's photo and choose their preferred style for AI photo generation.

Level 2 Data Flow Diagram: Manage Booking

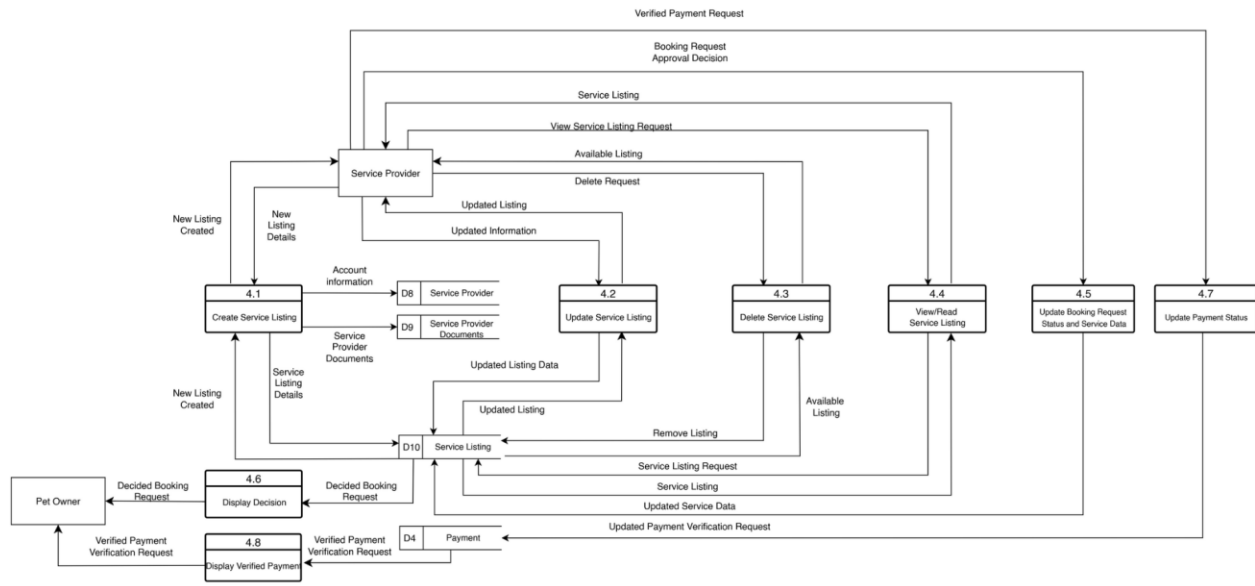


Figure 7 Level 2 Data Flow Diagram Manage Listing

The level 2 DFD of Manage Listing consists of Create, Modify, View, Delete, booking request, and payment verification of the Service Provider side. The first process is when the service provider Creates a service listing, and the data gets stored to datastores Service provider, service provider documents, and Service Listing then returns the new listing created. Second, the service provider updates a service listing information. The modified info also gets updated in the db and returns the listing with updated info. Third, the service provider deletes a listing and gets removed from the data store. After that it returns available listings only. Fourth, the service provider requests to view their service listing. Fifth, the service provider accepts or declines a booking request which gets updated in the service listing data store. Then the pet owner receives the decided booking request. Lastly, the service provider confirms or denies the payment verification request and gets updated in the datastore payment. Then the pet owner receives the verified payment request.

Level 2 Data Flow Diagram: Generate Report

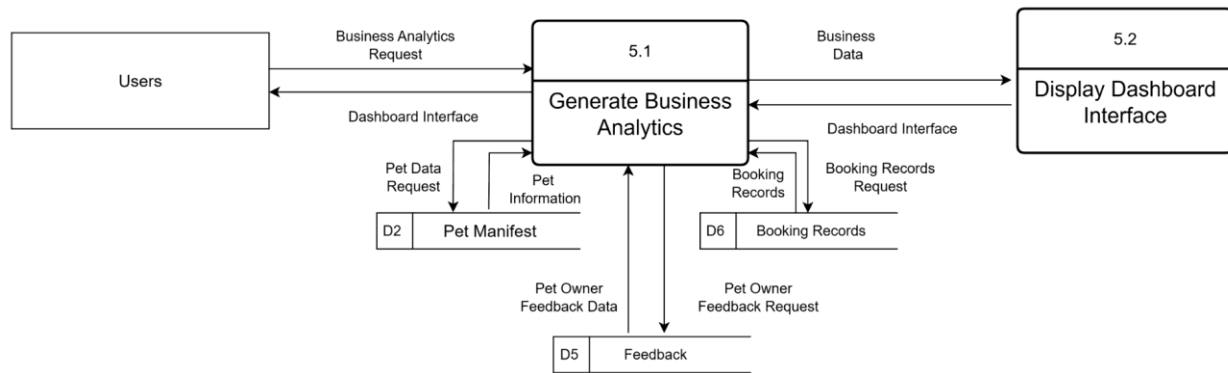


Figure 8 Level 2 Data Flow Generate Report

When the users send a Business Analytics Request, process 5.1 gathers data from the three data stores. After collecting all necessary data, the system produces the Business Data, which is then sent to process 5.2. This process displays the analytics and insights on the admin or service provider dashboard.

Level 2 Data Flow Diagram: Manage User Activity

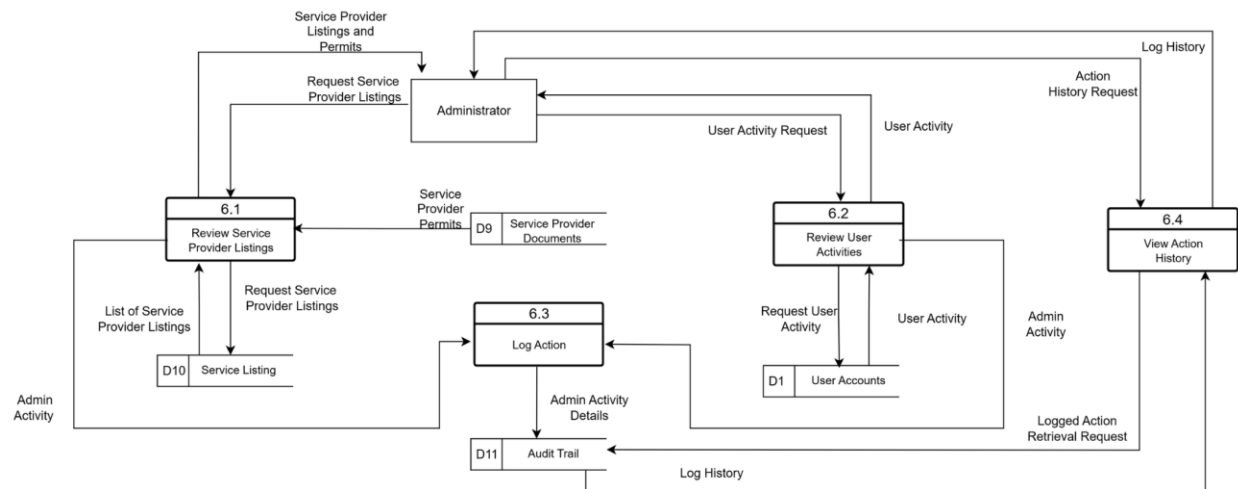


Figure 9 Level 2 Data Flow Diagram Manage User Activity

For the Level 2 DFD of moderate user activity, the administrator will send a request to review user activities then select the necessary action. Similar to moderating service provider listings, the administrator will send a request to review listings along with the documents then approve or reject.

Activity Diagrams with Swimlanes

An activity diagram with swimlane shows the whole process of a system. It represents a clear and concise visualization of the system's architecture and design and promotes understanding and collaboration.

Please click this link to view the Activity Diagram with Swimlanes [Activity Diagram with Swimlanes](#)

Test Cases

Test cases are a detailed set of instructions used to verify whether a specific functionality of a software application is working as expected. It defines the execution steps, input data, and the expected outcome to validate if the software behaves as expected.

Please click this link to view the Test Cases [Test Cases](#)

Entity Relationship Diagram

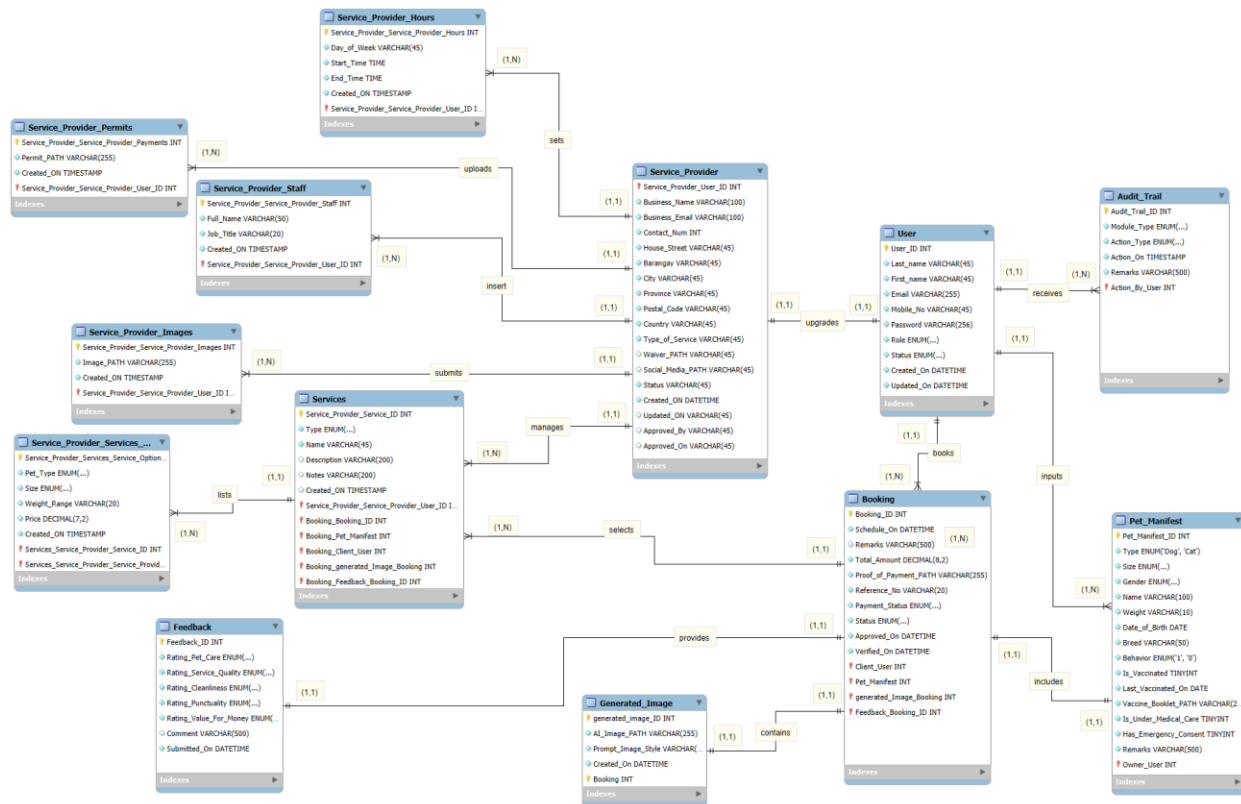


Figure 10 Entity Relationship Diagram

furlink's Entity-Relationship Diagram (ERD) illustrates the entities, attributes, relationships and cardinality of each data store within the system. It demonstrates how essential information—such as Pet Owner details, Pet Manifest, and Service Provider business information—is stored to achieve the system's full functionality.

Technology Stack

Frontend Framework

The team will be utilizing React.js with React Router, Cascading Style Sheet, and React Icons as part of the project's frontend framework.

Backend Framework

furlink will be built using Node.js, Express.js, and JSON Web Tokens or JWT.

Frontend and Backend Communication

The project will use REST API and Supabase Client for the communication of the frontend and backend.

Github APC-SoCIT project repository

Please click this link to view the Github APC-SoCIT [|URL Link|](#)

Openproject Team Site

Please click this link to view the Openproject Team Site [|URL Link|](#)

Cloud Hosted Site

Please click this link to view the cloud hosted site [|URL Link|](#)